



Variant Edge

Reproducible GRCh38 Rare & Genetic Disease SNV and CNV Annotation Pipeline

Diseases Covered

Fabry disease
Monilethrix
Androgen Insensitivity Syndrome
Neurofibromatosis Type 1

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All variant files are synthetic educational data and do not represent real patients.

1. Introduction

This assignment records the complete development, validation, automation, and interpretation workflow used for a reproducible GRCh38 rare-disease & genetic-disease annotation project.

The project was designed to process small sequence variants and copy-number variants as two separate branches. The small-variant branch accepts an unannotated VCF and produces a final VCF containing VEP, SnpEff, ClinVar, and SpliceAI annotations. The CNV branch accepts a four-column BED file and produces AnnotSV, ClassifyCNV, ISV-CNV, and combined-summary outputs.

Every tool was first tested manually in an Apptainer environment. The confirmed commands were then assembled into a disease-specific Bash pipeline and a batch runner for four diseases. The datasets are synthetic educational files and must not be presented as real patient results or used for diagnosis.

The documentation focuses on reproducibility: tool purpose, input, command, output, validation, errors, solutions, final file organization, meaningful screenshots, and GitHub links.

Important interpretation boundary: The sequence and CNV findings described here are synthetic educational targets. The report documents the pipeline and its outputs; it is not a clinical laboratory report.

2. Selected Diseases, Genes, Mechanisms, and Synthetic Targets

The four active diseases and their main disease-focused variants are summarized below. Disease mechanisms are included only to explain why the selected gene and variant are biologically relevant to the workflow.

Disease	Primary gene / locus	Inheritance	Mechanism	Synthetic SNV target	Synthetic CNV target
Fabry disease	GLA; chrX	X-linked	Deficient alpha-galactosidase A causes lysosomal accumulation of globotriaosylceramide and related substrates.	rs104894828; GLA c.902G>A, p.Arg301Gln; pathogenic missense	chrX:101393272-101408012 DEL; includes GLA; pathogenic by AnnotSV and ClassifyCNV
Monilethrix	KRT86 (target SNV); DSG4 (target CNV)	Usually autosomal dominant for keratin genes; autosomal recessive for DSG4	Hair keratin or desmosomal disruption weakens the hair-shaft cortex, producing fragile, beaded hair.	rs121909130; KRT86 c.1239G>T, p.Glu413Asp; pathogenic missense	chr18:31376776-31414979 DEL; includes DSG4; AnnotSV pathogenic, ClassifyCNV uncertain
Androgen Insensitivity Syndrome	AR; chrX	X-linked	Loss of androgen-receptor function causes reduced cellular response to androgens across a spectrum of complete, partial, or mild AIS.	rs1555997580; AR c.2359C>T, p.Arg787Ter; pathogenic stop-gained	chrX:67543352-67730619 DEL; includes AR; pathogenic by AnnotSV and ClassifyCNV
Neurofibromatosis Type 1	NF1; chr17q11.2	Autosomal dominant	Loss of neurofibromin removes negative regulation of RAS signaling and supports abnormal cellular proliferation.	rs267606599; NF1 c.3113+1G>A; pathogenic splice-donor variant; SpliceAI donor-loss 1.00	chr17:31094882-31382116 DEL; includes NF1 and neighboring genes; pathogenic by all three CNV tools

3. Project Objective and Final Workflow

The objective was to build a portable, reproducible, container-based workflow that can be run with a single meaningful disease name such as fabry, monilethrix, ais, or nf1.

```
bash pipeline/run_disease_pipeline.sh fabry
bash pipeline/run_all_diseases.sh
```

3.1 Small-variant branch

```
Unannotated SNV/indel VCF
-> bcftools norm
-> VEP 115 (offline cache 115)
-> SnpEff
-> ClinVar annotation
-> SpliceAI
-> final annotated VCF.gz
```

3.2 CNV branch

```
Unannotated CNV VCF
-> bcftools query conversion to 4-column BED
-> AnnotSV
-> ClassifyCNV
-> ISV-CNV with SHAP values
-> combined CNV summary TSV
```

4. Project Structure and Naming Convention

```
~/rare_disease_project/
├─ containers/      # Apptainer .sif files and .def recipes
├─ resources/      # FASTA, VEP cache, SnpEff, ClinVar, AnnotSV resources
├─ tools/          # ClassifyCNV and supporting software
├─ input/
│   ├── small_variants/  # unannotated disease VCF files
│   └─ cnv/             # CNV VCF and converted BED files
├─ pipeline/
│   ├── run_disease_pipeline.sh
│   └─ run_all_diseases.sh
├─ results/
│   ├── fabry/
│   ├── monilethrix/
│   ├── ais/
│   └─ nf1/
```

Meaningful base names were used consistently so the pipeline can discover paired inputs automatically:

Disease base name	SNV input	CNV BED input
fabry	fabry.small_variants.vcf	fabry.cnvs.bed
monilethrix	monilethrix.small_variants.vcf	monilethrix.cnvs.bed
ais	ais.small_variants.vcf	ais.cnvs.bed
nf1	nf1.small_variants.vcf	nf1.cnvs.bed

Figure 1. Project folder structure

```
(base) abdurrehman000@DESKTOP-0A9I4RJ:~/rare_disease_project$ cd ~/rare_disease_project

tree -d -L 1 \
-I '__pycache__|*_ClassifyCNV|sample_ClassifyCNV'
.
├── containers
├── input
├── pipeline
├── resources
├── results
└── tools

7 directories
```

5. Step 1 - Prepare the Project Folders

The project folders separate executable containers, reference resources, unannotated inputs, intermediate results, final outputs, and documentation.

```
mkdir -p ~/rare_disease_project/{containers,resources,input,pipeline,results,tools}
mkdir -p ~/rare_disease_project/resources/{reference,vep_cache,snpEff_data,clinvar,clingen,annovar_humandb,annotsv_annotations}
mkdir -p ~/rare_disease_project/input/{small_variants,cnv}
```

The `-p` option creates parent folders when needed and does not fail if a folder already exists. The `~` symbol represents the current user home directory.

6. Step 2 - Confirm Apptainer, Containers, and Resources

Apptainer was selected because the workflow must be portable to systems that provide Apptainer or Singularity. The project binds the host project root to `/project` inside each container.

```
apptainer --version
ls -lh ~/rare_disease_project/containers

apptainer exec --bind ~/rare_disease_project:/project ~/rare_disease_project/containers/core_tools.sif bash -lc 'bcftools --version | head -n 1; samtools --version | head -n 1; python3 --version'
```

Container	Purpose	Important content
core_tools.sif	General command-line utilities	bcftools, samtools, bgzip, tabix, Python and supporting tools
vep.sif	Sequence consequence annotation	VEP 115; must match cache 115
snpEff.sif	Independent functional consequence annotation	Snpeff and GRCh38 MANE database
spliceai.sif	Splice-effect prediction	SpliceAI command-line program
annotsv.sif	CNV annotation	AnnotSV 3.5.10
isv.sif	CNV machine-learning prediction	ISV 0.3.16 with pinned compatible scientific stack

Figure 2. Tool and container confirmation

```
/home/abdurrehman000/rare_disease_project/containers/annotsv.sif
/home/abdurrehman000/rare_disease_project/containers/core_tools.sif
/home/abdurrehman000/rare_disease_project/containers/isv.sif
/home/abdurrehman000/rare_disease_project/containers/snpeff.sif
/home/abdurrehman000/rare_disease_project/containers/spliceai.sif
/home/abdurrehman000/rare_disease_project/containers/vep.sif
/home/abdurrehman000/rare_disease_project/containers/vep_release116.sif
```

7. Step 3 - Prepare the GRCh38 Reference Genome

The reference FASTA is required by normalization, VEP, and SpliceAI. The .fai index allows rapid chromosome access.

```
cd ~/rare_disease_project/resources/reference
gunzip -c hg38.fa.gz > hg38.fa

apptainer exec --bind ~/rare_disease_project:/project ~/rare_disease_project/containers/core_tools.sif samtools faidx
/project/resources/reference/hg38.fa

ls -lh hg38.fa hg38.fai
```

Do not upload to GitHub: The full GRCh38 FASTA and index are external resources, not project source files. Document their source and expected paths instead.

8. Step 4 - Arrange and Validate the Unannotated Inputs

Each disease requires one unannotated small-variant VCF and one CNV file. CNV VCFs were converted into four-column BED files because AnnotSV, ClassifyCNV, and ISV-CNV were run from the same standardized CNV representation.

8.1 Convert CNV VCF to BED

```
cd ~/rare_disease_project/input/cnv

for vcf in *.cnvs.vcf
do
    bed="{vcf%.vcf}.bed"

    apptainer exec --bind ~/rare_disease_project:/project ~/rare_disease_project/containers/core_tools.sif bcftools query -f
'%CHROM\t%POS0\t%END\t%INFO/SVTYPE\n' "/project/input/cnv/$vcf" | awk 'BEGIN{OFS="\t"}
{
    if ($1 !~ /^chr/) $1="chr"$1
    if ($4=="DEL" || $4=="DUP")
        print $1,$2,$3,$4
}' > "$bed"
done
```

The BED columns are chromosome, zero-based start, end, and CNV type. Only DEL and DUP records are retained.

8.2 Validate SNV and CNV inputs

```
for disease in fabry monilethrix ais nf1
do
```

```
test -s "input/small_variants/${disease}.small_variants.vcf" || echo "Missing SNV: $disease"
test -s "input/cnv/${disease}.cnvs.bed" || echo "Missing CNV: $disease"
done
```

Windows metadata: Files ending in :Zone.Identifier are Windows metadata and were removed with `rm -f -- *:Zone.Identifier`. They are not biological inputs.

Figure 3. Validated disease input pairs

```
(base) abdurrehman000@DESKTOP-OA9I4RJ:~/rare_disease_project$ for d in fabry monilethrix ais nf1; do printf '%-14s ' "$d"; test -s
input/small_varia
nts/$d.small_variants.vcf && test -s input/cnv/$d.cnvs.bed && echo PASS || echo FAIL; done
fabry      PASS
monilethrix PASS
ais        PASS
nf1        PASS
```

9. Step 5 - Create the Disease-Specific Automated Script

The main script accepts one lowercase disease base name, validates the paired inputs, creates disease-specific result folders, and runs the SNV and CNV branches.

```
bash pipeline/run_disease_pipeline.sh fabry
```

```
#!/usr/bin/env bash
set -Eeuo pipefail
IFS=$'\n\t'

SCRIPT_DIR="$(cd -- "$(dirname -- "${BASH_SOURCE[0]}")" && pwd)"
PROJECT_ROOT="$(dirname "$SCRIPT_DIR")"
DISEASE="${1:-}"

SNV_INPUT="$PROJECT_ROOT/input/small_variants/${DISEASE}.small_variants.vcf"
CNV_INPUT="$PROJECT_ROOT/input/cnv/${DISEASE}.cnvs.bed"
OUTDIR="$PROJECT_ROOT/results/${DISEASE}"

mkdir -p "$OUTDIR"/{snv,cnv,work,logs,final}
```

`set -Eeuo pipefail` makes the pipeline stop on errors, undefined variables, and failures inside pipelines. This prevents a failed annotation stage from being hidden.

10. Step 6 - Normalize Small Variants with bcftools

Item	Description
Input	<disease>.small_variants.vcf
Purpose	Split multiallelic records, left-align variants, verify REF bases against GRCh38, and produce a standardized compressed VCF.
Key field/output	Normalized VCF.gz plus optional .tbi index
Output	results/<disease>/work/<disease>.normalized.vcf.gz

```
bcftools norm -f /project/resources/reference/hg38.fa -m -any -c x -Oz -o
/project/results/${DISEASE}/work/${DISEASE}.normalized.vcf.gz /project/input/small_variants/${DISEASE}.small_variants.vcf
```

The -c x option excludes records whose REF allele does not match the GRCh38 reference. In the synthetic datasets, some placeholder background variants were removed for this reason. The disease-focused variants remained present. This behavior must be disclosed rather than hidden.

11. Step 7 - Annotate with VEP 115

Item	Description
Input	Normalized VCF.gz
Purpose	Predict gene, transcript, coding, protein, canonical/MANE, HGVS, biotype, and variant-class consequences.
Important field	INFO/CSQ
Output	results/<disease>/snv/<disease>.vep.annotated.vcf

```
vep --input_file /project/results/${DISEASE}/work/${DISEASE}.normalized.vcf.gz --output_file
/project/results/${DISEASE}/snv/${DISEASE}.vep.annotated.vcf --format vcf --vcf --offline --cache --dir_cache /project/resources/vep_cache
--species homo_sapiens --assembly GRCh38 --fasta /project/resources/reference/hg38.fa --symbol --canonical --mane --hgvs --variant_class --
biotype --numbers --force_overwrite --no_stats --fork 4
```

12. Step 8 - Add SnpEff Consequences

Item	Description
Input	VEP-annotated VCF
Purpose	Add an independent transcript consequence annotation for comparison with VEP.
Important fields	INFO/ANN; optional LOF and NMD
Output	results/<disease>/snv/<disease>.vep.snpeff.annotated.vcf.gz

```
apptainer exec --bind "$PROJECT_ROOT:/project" "$SNPEFF_SIF" /opt/snpEff/exec/snpEff ann -noStats -canon -hgvs -dataDir
/project/resources/snpeff_data/data GRCh38.mane.1.2.ensembl "$CONTAINER_VEP_OUTPUT" | apptainer exec --bind
"$PROJECT_ROOT:/project" "$CORE_SIF" bzip -c > "$SNPEFF_OUTPUT"
```

13. Step 9 - Add ClinVar Clinical Annotations

Item	Description
Input	VEP + SnpEff VCF.gz
Purpose	Add known clinical significance, disease names, review status, HGVS, variant type, and gene identifiers for exact ClinVar matches.
Important fields	CLNSIG, CLNREVSTAT, CLNDN, CLNDISDB, CLNHGVS, CLNVC, GENEINFO
Output	results/<disease>/snv/<disease>.vep.snpeff.clinvar.annotated.vcf.gz

```
bcftools annotate -a /project/resources/clinvar/clinvar.chr.vcf.gz -c
INFO/CLNSIG,INFO/CLNREVSTAT,INFO/CLNDN,INFO/CLNDISDB,INFO/CLNHGVS,INFO/CLNVC,INFO/CLNVCISO,INFO/GENEINFO -Oz -o
"$CONTAINER_CLINVAR_OUTPUT" "$CONTAINER_SNPEFF_OUTPUT"
```

A variant with no ClinVar match is not automatically benign. It only means that the exact allele was not annotated by the selected ClinVar resource.

14. Step 10 - Predict Splice Effects with SpliceAI

Item	Description
Input	VEP + SnpEff + ClinVar VCF.gz
Purpose	Predict acceptor gain/loss and donor gain/loss effects and their relative positions.
Important field	INFO/SpliceAI: DS_AG, DS_AL, DS_DG, DS_DL and positions
Output	results/<disease>/snv/<disease>.vep.snpeff.clinvar.spliceai.vcf.gz

```
spliceai -I "$CONTAINER_CLINVAR_OUTPUT" -O "$CONTAINER_SPLICEAI_RAW" -R /project/resources/reference/hg38.fa -A grch38
```

The SpliceAI output is compressed and copied to the final folder as <disease>.final.small_variants.annotated.vcf.gz. A .tbi index is optional for submission but useful for fast regional queries.

Figure 4. Compact disease SNV evidence

```
(base) abdurrehman000@DESKTOP-0A9I4RJ:~/rare_disease_project$ aptainer exec --bind ~/rare_disease_project:/project
~/rare_disease_project/container
s/core_tools.sif bcftools query -f
'%CHROM\t%POS\t%ID\t%REF\t%ALT\t%INFO/CLNSIG\t%INFO/CLNREVSTAT\t%INFO/CLNDN\t%INFO/SpliceAI\n' /project/results/a
is/final/ais.final.small_variants.annotated.vcf.gz | grep rs1555997580
chrX 67721873 rs1555997580 C T Pathogenic criteria_provided_multiple_submitters_no_conflicts Androgen_resistance_
syndrome|not_provided|Kennedy_disease T|AR|0.00|0.00|0.00|22|-40|-41|-30
```

15. Step 11 - Annotate CNVs with AnnotSV

Item	Description
Input	Four-column CNV BED
Purpose	Annotate genes, transcripts, cytobands, dosage-sensitive regions, pathogenic/benign overlaps, known disease associations, ranking criteria, and an AnnotSV ACMG class.
Output	results/<disease>/cnv/<disease>.AnnotSV.tsv

```
AnnotSV -SVinputFile /project/input/cnv/${DISEASE}.cnvs.bed -outputFile /project/results/${DISEASE}/cnv/${DISEASE}.AnnotSV.tsv -
annotationsDir /project/resources/annotsv_annotations/AnnotSV_annotations -genomeBuild GRCh38 -svtBEDcol 4 -annotationMode both -
overwrite 1
```

16. Step 12 - Classify CNVs with ClassifyCNV

Item	Description
Input	The same four-column CNV BED used by AnnotSV
Purpose	Apply ACMG/ClinGen CNV evidence scoring and report the total score, classification, dosage-sensitive genes, and section-level evidence.
Output	results/<disease>/cnv/<disease>.ClassifyCNV.Scoresheet.txt

```
python3 ClassifyCNV.py --infile /project/input/cnv/${DISEASE}.cnvs.bed --GenomeBuild hg38 --cores 4 --precise --outdir
${DISEASE}_ClassifyCNV
```


Parallel interpretation: AnnotSV output is not used as ClassifyCNV input. Both tools independently analyze the same CNV BED and are compared afterward.

17. Step 13 - Run ISV-CNV and SHAP Analysis

Item	Description
Input	Header-containing CNV BED: chromosome, start, end, cnv_type
Purpose	Generate an independent machine-learning pathogenicity probability and SHAP contributions showing which feature groups influenced the prediction.
Output	results/<disease>/cnv/<disease>.ISV_with_SHAP.tsv

```
{
  printf "chromosome\tstart\tend\tcnv_type\n"
  cat "$CNV_INPUT"
} > "$ISV_INPUT"

results = isv.isv(cnvs, proba=True, shap=True, threshold=0.95)
results.to_csv(output_file, sep="\t", index=False)
```

The ISV package was run through its Python API because the installed package did not create an isv executable in PATH. The probability is supporting computational evidence and does not replace curated evidence.

18. Step 14 - Combine the Three CNV Outputs

A pandas script inside isv.sif retains one full AnnotSV row per CNV, restores the original zero-based BED start, and merges AnnotSV, ClassifyCNV, and ISV by chromosome, start, end, and type.

Combined column	Meaning
Gene_count	Number of genes reported in the full AnnotSV row
AnnotSV_ranking_score	AnnotSV evidence-based ranking value
AnnotSV_ACMG_class / classification	AnnotSV class 1-5 and corresponding label
ClassifyCNV_classification / total_score	Independent ACMG/ClinGen classification and score
ISV_probability	Independent machine-learning pathogenicity probability
ISV_threshold_0.95	Whether the probability met the selected 0.95 threshold

Final output: results/<disease>/final/<disease>.final.cnv.summary.tsv

Figure 5. Compact combined CNV summary

```
(base) abdurrehman000@DESKTOP-0A914RJ:~/rare_disease_project$ awk -F '\t' 'NR==1 || ($1=="chr17" && $2==31094882)'
~/rare_disease_project/results/n
f1/final/nf1.final.cnv.summary.tsv | column -t -s $'\t'
Chromosome Start End Type Gene_count AnnotSV_ranking_score AnnotSV_ACMG_class AnnotSV_classification ClassifyCNV_classification
ssifyCNV_total_score ISV_probability ISV_threshold_0.95
chr17 31094882 31382116 DEL 1 1.0 5 Pathogenic Pathogenic 1.0
0 0.990364134311676 Above threshold
```

19. Step 15 - Final Outputs, Manifest, Logs, and Batch Execution

Final file	Purpose
<disease>.final.small_variants.annotated.vcf.gz	Final VEP + SnpEff + ClinVar + SpliceAI small-variant result
<disease>.final.small_variants.annotated.vcf.gz.tbi	Optional VCF index for rapid queries
<disease>.final.cnv.summary.tsv	Combined AnnotSV + ClassifyCNV + ISV-CNV summary
<disease>.pipeline_outputs.txt	Manifest listing inputs, outputs, record counts, tools, and log path
<disease>.pipeline.log	Detailed run log for troubleshooting and reproducibility

```
DISEASES=(fabry monilethrix ais nf1)
for disease in "${DISEASES[@]}"
do
  bash "$PIPELINE" "$disease"
done
```

Figure 6. Successful batch completion

```
=====
PIPELINE COMPLETED SUCCESSFULLY
=====

Disease: nf1
Final SNV: /home/abdurrehman000/rare_disease_project/results/nf1/final/nf1.final.small_variants.annotated.vcf.gz
Final CNV: /home/abdurrehman000/rare_disease_project/results/nf1/final/nf1.final.cnv.summary.tsv
Manifest: /home/abdurrehman000/rare_disease_project/results/nf1/final/nf1.pipeline_outputs.txt
Log: /home/abdurrehman000/rare_disease_project/results/nf1/logs/nf1.pipeline.log
PASS: nf1 completed successfully
```

20. Disease-Specific Evidence

Use only two core evidence screenshots per disease: one compact SNV record and one compact CNV summary row. This produces eight understandable disease screenshots. Additional screenshots are optional only when they explain a specific error or tool disagreement.

20.1 Fabry disease

Evidence branch	Meaningful result
SNV	GLA c.902G>A (p.Arg301Gln), pathogenic missense
CNV	GLA-containing deletion; AnnotSV pathogenic; ClassifyCNV pathogenic; ISV 0.897, below 0.95

Figure 7. Fabry disease - disease-focused SNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ apptainer exec --bind ~/rare_disease_project:/project ~:/rare_disease_project/container
s/core_tools.sif bcftools query -f
'%CHROM\t%POS\t%ID\t%REF\t%ALT\t%INFO/CLNSIG\t%INFO/CLNREVSTAT\t%INFO/CLNDN\t%INFO/SpliceAI\n'
/project/results/fabry/final/fabry.final.small_variants.annotated.vcf.gz | grep rs104894828
chrX 101398467 rs104894828 C T Pathogenic criteria_provided_multiple_submitters_no_confl
icts Cardiovascular_phenotype|not_provided|Fabry_disease|Fabry_disease_cardiac_variant|Primary_familial_hypertrophic
_cardiomyopathy T|RPL36A-HNRNP2|0.00|0.00|0.00|0.00|13|-28|0|4,T|GLA|0.00|0.00|0.00|0.00|-35|-10|-35|-43
```

Figure 8. Fabry disease - disease-focused CNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ awk -F '\t' 'NR==1 || ($1=="chrX" && $2==101393272)' ~/rare_disease_project/results/fabry/final/fabry.final.cnv.summary.tsv | column -t -s $'\t'
Chromosome Start End Type Gene_count AnnotSV_ranking_score AnnotSV_ACMG_class AnnotSV_classification ClassifyCNV_classification ClassifyCNV_total_score ISV_probability ISV_threshold_0.95
chrX 101393272 101408012 DEL 3 1.0 5 Pathogenic Pathogenic
1.00 0.8970537781715393 Below threshold
```

GitHub links for this disease:

Unannotated SNV VCF: [Link](#)

GPT-annotated educational SNV VCF (if retained): [Link](#)

Pipeline-annotated final SNV VCF: [Link](#)

Unannotated CNV VCF: [Link](#)

Converted CNV BED: [Link](#)

Combined CNV summary: [Link](#)

20.2 Monilethrix

Evidence branch	Meaningful result
SNV	KRT86 c.1239G>T (p.Glu413Asp), pathogenic missense
CNV	DSG4-containing deletion; AnnotSV pathogenic; ClassifyCNV uncertain; ISV 0.651

Figure 9. Monilethrix - disease-focused SNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ aptainer exec --bind ~/rare_disease_project:/project ~/rare_disease_project/containers/core_tools.sif bcftools query -f '%CHROM\t%POS\t%ID\t%REF\t%ALT\t%INFO/CLNSIG\t%INFO/CLNREVSTAT\t%INFO/CLNDN\t%INFO/SpliceAI\n' /project/results/monilethrix/final/monilethrix.final.small_variants.annotated.vcf.gz | grep rs121909130
chr12 52306272 rs121909130 G T Pathogenic no_assertion_criteria_provided Monilethrix|not_provided T
|KRT86|0.00|0.00|0.00|0.00|-30|-50|-2|-5
```

Figure 10. Monilethrix - disease-focused CNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ awk -F '\t' 'NR==1 || ($1=="chr18" && $2==31376776)' ~/rare_disease_project/results/monilethrix/final/monilethrix.final.cnv.summary.tsv | column -t -s $'\t'
Chromosome Start End Type Gene_count AnnotSV_ranking_score AnnotSV_ACMG_class AnnotSV_classification ClassifyCNV_classification ClassifyCNV_total_score ISV_probability ISV_threshold_0.95
chr18 31376776 31414979 DEL 2 1.0 5 Pathogenic Uncertain signif
icance 0.00 0.6507948040962219 Below threshold
```

GitHub links for this disease:

Unannotated SNV VCF: [Link](#)

GPT-annotated educational SNV VCF: [Link](#)

Pipeline-annotated final SNV VCF: [Link](#)

Unannotated CNV VCF: [Link](#)

Converted CNV BED: [Link](#)

Combined CNV summary: [Link](#)

20.3 Androgen Insensitivity Syndrome

Evidence branch	Meaningful result
SNV	AR c.2359C>T (p.Arg787Ter), pathogenic stop-gained
CNV	AR deletion; AnnotSV pathogenic; ClassifyCNV pathogenic; ISV 0.541

Figure 11. Androgen Insensitivity Syndrome - disease-focused SNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ aptainer exec --bind ~/rare_disease_project:/project ~/rare_disease_project/containers
/core_tools.sif bcftools query -f '%CHROM\t%POS\t%ID\t%REF\t%ALT\t%INFO/CLNSIG\t%INFO/CLNREVSTAT\t%INFO/CLNDN\t%INFO/SpliceAI\n' /project/results/ais/final/ais.final.small_variants.annotated.vcf.gz | grep rs1555997580
chrX 67721873 rs1555997580 C T Pathogenic criteria_provided,multiple_submitters,no_confli
cts Androgen_resistance_syndrome|not_provided|Kennedy_disease T|AR|0.00|0.00|0.00|22|-40|-41|-30
```

Figure 12. Androgen Insensitivity Syndrome - disease-focused CNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ awk -F '\t' 'NR==1 || ($1=="chrX" && $2==67543352)' ~/rare_disease_project/results/ais/
final/ais.final.cnv.summary.tsv | column -t -s $'\t'
Chromosome Start End Type Gene_count AnnotSV_ranking_score AnnotSV_ACMG_class AnnotSV_classification Clas
sifyCNV_classification ClassifyCNV_total_score ISV_probability ISV_threshold_0.95
chrX 67543352 67730619 DEL 1 1.0 5 Pathogenic Path
ogenic 1.00 0.5410510897636414 Below threshold
```

GitHub links for this disease:

Unannotated SNV VCF: [Link](#)

GPT-annotated educational SNV VCF: [Link](#)

Pipeline-annotated final SNV VCF: [Link](#)

Unannotated CNV VCF: [Link](#)

Converted CNV BED: [Link](#)

Combined CNV summary: [Link](#)

20.4 Neurofibromatosis Type 1

Evidence branch	Meaningful result
SNV	NF1 c.3113+1G>A, pathogenic splice-donor; SpliceAI donor loss 1.00
CNV	NF1-containing deletion; AnnotSV pathogenic; ClassifyCNV pathogenic; ISV 0.990, above threshold

Figure 13. Neurofibromatosis Type 1 - disease-focused SNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ aptainer exec --bind ~/rare_disease_project:/project ~/rare_disease_project/containers
/core_tools.sif bcftools query -f '%CHROM\t%POS\t%ID\t%REF\t%ALT\t%INFO/CLNSIG\t%INFO/CLNREVSTAT\t%INFO/CLNDN\t%INFO/SpliceAI\n' /project/results/nf1/final/nf1.final.small_variants.annotated.vcf.gz | grep rs267606599
chr17 31230383 rs267606599 G A Pathogenic criteria_provided,multiple_submitters,no_confli
cts Cardiovascular_phenotype|Hereditary_cancer-predisposing_syndrome|Tibial_pseudarthrosis|Neurofibromatosis_type_1|
NF1-related_disorder|not_provided A|NF1|0.00|0.00|0.06|1.00|20|-13|20|-1
```

Figure 14. Neurofibromatosis Type 1 - disease-focused CNV

```
abdurrehman000@DESKTOP-OA9I4RJ:~$ awk -F '\t' 'NR==1 || ($1=="chr17" && $2==31094882)' ~/rare_disease_project/results/nf1
```

```
/final/nf1.final.cnv.summary.tsv | column -t -s '$'\t'
```

Chromosome	Start	End	Type	Gene_count	AnnotSV_ranking_score	AnnotSV_ACMG_class	AnnotSV_classification	Class
chr17	31094882	31382116	DEL	1	1.0	5	Pathogenic	Pathogenic
ogenic	1.00	0.990364134311676					Above threshold	

GitHub links for this disease:

Unannotated SNV VCF: [Link](#)

GPT-annotated educational SNV VCF: [Link](#)

Pipeline-annotated final SNV VCF: [Link](#)

Unannotated CNV VCF: [Link](#)

Converted CNV BED: [Link](#)

Combined CNV summary: [Link](#)

22. WinterVar Results:

Figure 14. Fabry

Database version:hg38_update_2025

You searched by dbSNP ID with **rs104894828**

Show/hide columns

Restore columns

Copy to clipboard

Download result as CSV

Search:

Chr	Position	Ref	Alt	Gene (refGene)	Intervar	ExonicFunc (refGene)	SNP	Transcripts (Ref)	MAF in gnomAD_ALL(genome)	Disease in OrphaNet
X	101398467	C	A	GLA	Likely pathogenic (Details&Adjust)	nonsynonymous SNV	rs104894828(details of MAF)	NM_000169 p.R301L	(show in 7 POPs)	324
X	101398467	C	T	GLA	Likely pathogenic (Details&Adjust)	nonsynonymous SNV	rs104894828(details of MAF)	NM_000169 p.R301Q	(show in 7 POPs)	324

Showing 1 to 2 of 2 entries

Previous

1

Next

Database version:hg38_update_2025

You searched by dbSNP ID with **rs104894828**

Show/hide columns

Restore columns

Copy to clipboard

Download result as CSV

Search:

	ExonicFunc (refGene)	SNP	Transcripts (Ref)	MAF in gnomAD_ALL(genome)	Disease in OrphaNet	OMIM	Phenotype_MIM	Interpro_domain	OrphaNo.
nic it	nonsynonymous SNV	rs104894828(details of MAF)	NM_000169 p.R301L	(show in 7 POPs)	324	300644	301500	..	ORPHA324
nic it	nonsynonymous SNV	rs104894828(details of MAF)	NM_000169 p.R301Q	(show in 7 POPs)	324	300644	301500	..	ORPHA324

Showing 1 to 2 of 2 entries

Previous

1

Next

Figure 14. Monilethrix

ExonicFunc (refGene)	SNP	Transcripts (Ref)	MAF in gnomAD_ALL(genome)	Disease in OrphaNet	OMIM	Phenotype_MIM	Interpro_domain	OrphaNo.
nonsynonymous SNV	rs121909130(details of MAF)	NM_001320198 p.E413D	(show in 7 POPs)	573	601928	158000	Intermediate filament, rod domain Intermediate filament, rod domain Intermediate filament, rod domain Intermediate filament, rod domain Intermediate filament, rod domain	ORPHA573

Chr	Position	Ref	Alt	Gene (refGene)	Interval	ExonicFunc (refGene)	SNP	Transcripts (Ref)	MAF in gnomAD_ALL(genome)	Disease in OrphaNet
12	52306272	G	T	KRT86	Uncertain significance (Details&Adjust)	nonsynonymous SNV	rs121909130(details of MAF)	NM_001320198 p.E413D	(show in 7 POPs)	573

WinterVar does not have the data of variants of AIS and NF1 related diseases selected in this pipeline.

24. Interpretation and Reporting Limitations

- All files are synthetic educational datasets and do not represent real patients.
- The pipeline records computational and database annotations; it does not independently establish a clinical diagnosis.
- Some background variants were removed because their REF bases did not match GRCh38 during bcftools norm -c x.
- A ClinVar match is useful supporting evidence, but classifications require date-stamped review in real clinical work.
- A low or zero SpliceAI score does not make a non-splicing variant benign.
- ISV probability is supporting computational evidence and should not override curated evidence.
- CNV breakpoint accuracy, zygoty, phase, phenotype matching, segregation, and orthogonal confirmation were not available.
- The report should distinguish disease-focused synthetic targets from background variants and from real clinical conclusions.

25. Conclusion

The completed workflow demonstrates a reproducible GRCh38 annotation system with separate sequence-variant and CNV branches. The manual validation phase identified software-version, executable-path, dependency, input-format, and coordinate-consistency issues before automation. The final pipeline now accepts a disease base name, runs the confirmed tools,

validates each stage, writes disease-specific outputs, and supports batch execution across Fabry disease, Monilethrix, AIS, and NF1.